Pablo Francisco Coral

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EXPERIENCE & LEADERSHIP

Teaching Assistant **Aug 2024 – Present**

Design and Manufacturing Lab, UF *Gainesville, FL*

- Teach and run the lab for groups of four students through a semester-long build of compressed-air engines, fully owning their hands-on instruction on lathes, mills, bandsaws, MIG welders, and sheet metal equipment from machining fundamentals to final assembly.
- Mentor each team through design, fabrication, and assembly; review engineering drawings and reports for Design for Manufacturability and Assembly (DFMA) and grade assignments with feedback that drives improvement across the semester.
- Train students on shop safety at the start of every semester and enforce safety protocols during lab sessions; CNC-machine engine components and replacement parts for shop equipment to keep projects and machines running.
- Earned the ACE CNC certificate by programming, CNC-machining, and assembling 5 air-engine components in Fusion 360 on a Haas VF-3 mill.

Power Plant and Mechanical Systems Engineering Intern **Jan 2025 – May 2025**

Piper Aircraft Inc. *Vero Beach, FL*

- Self-taught heat transfer to develop a delay-differential model of aircraft brake rotor temperature from ~47,000 airspeed and pressure sensor readings; numerically solved with a 4th-order Runge-Kutta solver and optimized state vector with mini-batch stochastic gradient descent.
- Built a Python sensor-data pipeline with CSV ingestion, Savitzky-Golay smoothing, and Akima interpolation, as well as a weighted decision-table classifier that used airspeed, brake pressure, and altitude to identify brake rotor heating and cooling conditions for binning a large dataset.
- Developed three aircraft configurations in Microsoft Flight Simulator for VR-based pilot evaluation, configured flight dynamics in XML, and tuned pitch and roll response from pilot feedback; wrote a 14-page build guide for implementation on a custom-built flight simulator for future aircraft.
- Built a Python tool to quantify how aircraft nose length affects pilot visibility, using CAD geometry and trigonometric analysis across 0 to 19 inches of extension to find the pitch-down needed to keep baseline visibility and inform configuration decisions.
- Reverse-engineered an aftermarket oleo strut using precise metrology techniques and tools, taking over 200 measurements over 2 weeks to fully model the device in Siemens NX; briefed a team of 7 engineers on the mechanism of the strut to better understand how to design an in-house strut.

Co-Founder and Chief Technology Officer **Dec 2025 – Aug 2026**

Knead to Invest *Gainesville, FL*

- Architected and built the Strategy Playground in React and TypeScript, a visual trading-strategy builder where users drag Blockly blocks that a custom abstract syntax tree (AST) translator converts into Python, run one-click Backtrader backtests, and read results on TradingView charts with buy/sell markers, indicator overlays, Sharpe ratio, and max drawdown benchmarked against the S&P 500.
- Built a working prototype version of each backend service, including the backtesting engine behind a Flask API gateway, polished each service once it worked at a basic level by delegating to our engineers, and partnered with the lead engineer to turn the system into a full microservices backend.
- Designed a live-trading architecture with per-strategy subprocesses, threaded market-data delivery, and a rate-limited order scheduler while keeping broker credentials outside strategy code and respecting market data provider API limits.
- Hired, led and mentored 5+ engineers, fully architecting the recruiting and technical interview process, successfully obtaining talent. Co-developed the pitch deck and business plan and presented as part of the **2026 GatorTank pitch competition, winning First Place and a $1,200 grant.**

Co-Founder and Sole Developer **Jun 2026 – Present**

Level: Calisthenics for All *Gainesville, FL*

- Built and launched a subscription calisthenics iOS App with React Native & Expo plus TypeScript and a Supabase PostgreSQL backend, all based on a co-founder's training system proven in a 700+ member club. Created a custom beta tester management software and ran a 28-day, 30+ user beta test.
- Implemented RevenueCat entitlement verification and centralized access control in Supabase to gate premium features and manage paid and beta access.

INVOLVEMENT

Bike Bros, *Partner and Chief Operations Officer* **May 2025 – Present**

- Built a network of 4+ mechanics to source and repair high-quality vintage bicycles for resale at affordable prices.
- Sold 30+ bikes across the Gainesville area, handling sales, marketing & outreach, and accounting & finances to ensure profitability and product quality.

Florida Rocket Lab, *Manufacturing and Fluids Engineer* **Aug 2023 – Jan 2025**

- Automated P&ID pressure loss verification by developing a custom Python script, reducing manual calculation time by ~4 hours for faster iteration.
- Prototyped and delivered a mass-flow-rate test stand scale, accurate to ±0.01g to aid with pressure vessel mass flow rate testing, utilizing a quarter-Wheatstone bridge with an ADC and Arduino for data collection, and CoolTerm for serial logging and collection into Excel for plotting and analysis.
- Machined 12+ precision parts for a thrust-vectoring gimbal system using a manual lathe and mill, creating detailed manufacturing guides in Jira.

Classical and Collaborative Pianist **Sep 2022 – Present**

- Collaborate with UF faculty, vocalists, instrumentalists, and composers for recitals, juries, and concerts; direct weekly worship music at a local Church with 20+ regular attendees while constantly learning new repertoire and **perform solo and chamber programs for audiences of up to 3,000.**

RELEVANT SKILLS

Programming | Modeling: Python, TypeScript, SQL, MATLAB, C++ (Arduino) | Numerical Methods, Signal Processing, Differential Equations

Software | Data: React, React Native, Expo, Flask, PostgreSQL, Backtrader, pandas, NumPy, SciPy, TensorFlow | Claude Code, Codex, Cursor

Cloud | Development Tools: AWS, Supabase, Render, Vercel, Resend | Git, GitHub, Jira, Confluence, RevenueCat, ClickUp

Design | Manufacturing: Siemens NX, SolidWorks, Fusion 360 CAD/CAM, Blender | ACE CNC Certified

EDUCATION

B.S. in Aerospace Engineering, B.M. in Piano Performance **Aug 2022 – May 2027**

University of Florida, Gainesville, FL GPA: 3.52